QP CODE: 26900121

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA(Honours)/MGU-BCA(Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC)–MG5DSCCND300-MATERNAL NUTRITION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S),Interest (I) and Appreciation(Ap)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions
Answer all questions
Each question carries **1** mark

1. Option C
2. Option A
3. Option A
4. Option A
5. Option D
6. Option C

Part B

Short Answer Type Questions.
Answer any **2** questions.
Each question carries **2** marks.

7. List the major sources of lead exposure during pregnancy.[K]/[CO2]

- Lead-based paints and household dust.
- Contaminated drinking water from lead pipes
- Industrial emissions from factories and smelting.
- Traditional medicines, cosmetics, and contaminated soil. **(0.5 mark for each point)**

8. Define the term Human Milk Bank. [K]/[CO3]

A Human Milk Bank is a service that collects,screens,processes,stores and distributes donated human breast milk to infants who cannot receive their own mother's milk.Donor milk is mainly used for premature,low birth weight or sick infants**.(1 mark for each point)**

9. Explain the significance of iodine during foetal growth.[U]/[CO2]

- Essential for thyroid hormone production and foetal growth.
- Supports brain development and prevents intellectual disability. (1mark for each point)

10. Analyze how maternal stress may interfere with the let-down reflex. [An]/[CO3]

Maternal Stress can inhibit the release of oxytocin, the hormone responsible for milk ejection. This can delay or reduce the efficiency of milk let-down during breastfeeding.**(1 mark for each point)**

Part C

Short Essay Type Questions.
Answer any **4** questions.
Each question carries **5** marks.

11.Describe the haematological changes during pregnancy [U]/[CO1]

- Blood volume: Increases by 30–50%.
- Physiological anaemia: Plasma volume increases more than RBC mass.
- RBC production: Increases, with higher iron, folic acid and vitamin B12 requirements.
- WBC and platelets: WBC count increases; platelet count may decrease slightly.
- Blood coagulation: Clotting factors increase, raising the risk of thrombosis**.(1 mark for each point)**

12.Compare the effects of adequate and inadequate nutrition on female reproductive health [An]/[CO2] (1 mark for each point)

Adequate Nutrition	**Inadequate Nutrition**
1. Maintains normal menstrual cycle.	Causes menstrual irregularities.
2. Supports normal ovulation and fertility.	May cause anovulation and infertility.
3. Maintains hormonal balance.	Causes hormonal imbalance.
4. Supports healthy pregnancy and fetal growth.	Increases risk of miscarriage and fetal growth restriction.
5. Prevents nutritional deficiencies and anaemia.	Increases risk of anaemia and nutritional deficiencies.

13.Explain how Breast milk is expressed with a Breast Pump.[U]/[CO3]
Steps:

- **Wash hands:** Wash hands thoroughly with soap and water and ensure the breast pump and its parts are clean.
- **Prepare the breast:** Sit comfortably and gently massage the breasts to stimulate milk flow.
- **Position the pump:** Place the breast shield over the nipple, ensuring the nipple is centred.
- **Express the milk:** Switch on the electric pump or operate the manual pump. Start with gentle suction and gradually increase to a comfortable level until milk flows.

- **Collect and store:** Express milk from both breasts, pour it into a clean, sterile container, label it with the date and time, and store it safely according to breast milk storage guidelines. **(5 mark)**

14.Explain how gastrointestinal changes help meet the nutritional demands of pregnancy[An]/[CO1]

- Decreased gastrointestinal motility: Progesterone slows intestinal movement, allowing more time for nutrient and water absorption.
- Nausea and vomiting: Hormonal changes, mainly increased hCG and estrogen, may affect food intake, especially during the first trimester.
- Heartburn: Relaxation of the lower oesophageal sphincter causes acid reflux.
- Constipation: Reduced intestinal motility and increased water absorption cause constipation.
- Increased appetite: Higher energy and nutrient requirements support fetal growth and maternal health.
- Displacement of stomach and intestines: The enlarging uterus compresses the digestive organs, causing early fullness and reduced meal capacity. **(1 mark for each point)**

15. Discuss the factors influencing foetal growth and development. [K][CO2]

- Genetic factors: Determine foetal size, growth and organ development.
- Maternal nutrition: Nutrient deficiencies may cause low birth weight and IUGR.
- Placental function: Supplies oxygen and nutrients to the foetus.
- Hormonal regulation: Hormones control foetal growth and organ maturation.
- Intrauterine environment: Maternal health, infections, smoking and alcohol influence foetal development**. (1 mark for each point)**

16. Explain the role of hormones during pregnancy that help in supporting effective breast feeding .[U][CO3]

Role of Hormones during Pregnancy

1) Prolactin-Stimulates the development of mammary glands and milk synthesis. Its level increases during pregnancy. Actual milk secretion is inhibited during pregnancy due to high estrogen and progesterone levels. **(1.5 Mark)**

2)Estrogen-Promotes growth of ducts in the mammary glands**,**Increases breast size during pregnancy. **(1.5 Mark)**

3)Progesterone-Helps in the development of alveoli and lobules of the breast,Prevents full milk secretion during pregnancy. **(1 Mark)**

4)Human Placental Lactogen (hPL)-Supports mammary gland growth and preparation for lactation . **(1 Mark)**

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Compare and contract the causes,symptoms,prevention and health consequences of common nutritional deficiencies in women of reproductive age.[An]/[CO2] (2 mark for each point)

Deficiency	Causes	Symptoms	Prevention	Health Consequences
Iron	Poor intake, menstrual blood loss, increased requirements.	Fatigue, weakness, pallor, breathlessness.	Iron-rich foods, vitamin C, iron supplements.	Anaemia, preterm birth, low birth weight.
Folic acid	Low intake of folate-rich foods, increased pregnancy requirements.	Fatigue, weakness, megaloblastic anaemia.	Green leafy vegetables, legumes, folic acid supplements.	Neural tube defects, poor fetal development.
Calcium	Inadequate intake, poor absorption.	Muscle cramps, weak bones, numbness.	Milk, curd, ragi, green leafy vegetables, supplements if needed.	Reduced bone strength, poor fetal bone development.
Vitamin D	Inadequate sunlight exposure, poor dietary intake.	Bone pain, muscle weakness, fatigue.	Sunlight exposure, fatty fish, fortified foods, supplements if needed.	Poor fetal bone development, maternal bone problems.
Vitamin B12	Low intake of animal foods, poor absorption.	Fatigue, megaloblastic anaemia, numbness and tingling.	Meat, fish, eggs, dairy products, fortified foods and supplements.	Neurological problems, impaired fetal growth and development.

18. Compare the nutritional requirements of a lactating mother with those of a non-lactating woman and explain the reasons for the differences. [An]/[CO3]

Nutritional Requirements

1) **Energy**
   A Non-lactating woman- 1,660–2,720 kcal/day, lactating woman- an additional 600 kcal during 0-6 months and additional 520 kcal during 7-12 months.Extra energy-breast milk production**.(1 Mark)**
2) **Protein**
   A Non-lactating woman- 46gm/day, lactating woman - an additional 17gm during 0-6 months and additional 13 gm during 7-12 months.Extra protein- Synthesis of breast milk and repair of maternal tissues. **(1 Mark)**
3) **Calcium**
   A Non-lactating woman- 1000mg/day, lactating woman-1200 mg during 0-6 months and 7-12 months. Extra calcium supports milk production and maintains maternal bone health. (1 Mark)

4) **Iron**
   A Non-lactating woman- 29mg/day, lactating woman-23 mg during 0-6 months and 7-12 months. Menstrual blood loss is often reduced during lactation, lowering iron requirements.(1 Mark)
5) **Vitamin A**
   A Non-lactating woman- 840µgm/day, lactating woman- 950 µgm during 0-6 months and 7-12 months. Required for the baby's growth, vision and immunity and for maintaining vitamin A in breast milk. (1 Mark)
6) **Vitamin C**
   A Non-lactating woman- 65mg/day, lactating woman- additional 50mg/day during 0-6 months and 7-12 months. Supports tissue repair and provides vitamin C through breast milk. (1 Mark)
7) **Folate**
   A Non-lactating woman- 220 µgm/day, lactating woman- 330 µgm/day during 0-6 months and 7-12 months. Supports maternal health and the nutritional quality of breast milk. (1 Mark)
8) **Vitamin B12**
   A Non-lactating woman-2.2 µgm /day,a lactating woman- additional 1 µgm /day during 0-6 months and 7-12 months. Essential for the baby's nervous system development and red blood cell formation. (1 Mark)
9) **Iodine**
   A Non-lactating woman-140 µgm /day,a lactating woman- 280 µgm /day during 0-6 months and 7-12 months. Essential for the baby's thyroid hormone production, brain development and normal growth. (1 Mark)
10) **Zinc**
    A Non-lactating woman-13.2mg/day, lactating woman- 14.1gm/day during 0-6 months and 7-12 months. Supports the baby's growth, immune function, also helps in maternal tissue repair. (1 Mark)

19. Describe the functional anatomy and functions of the fallopian tubes with a neat labelled diagram. [U]/[CO1]

- **Functional Anatomy**

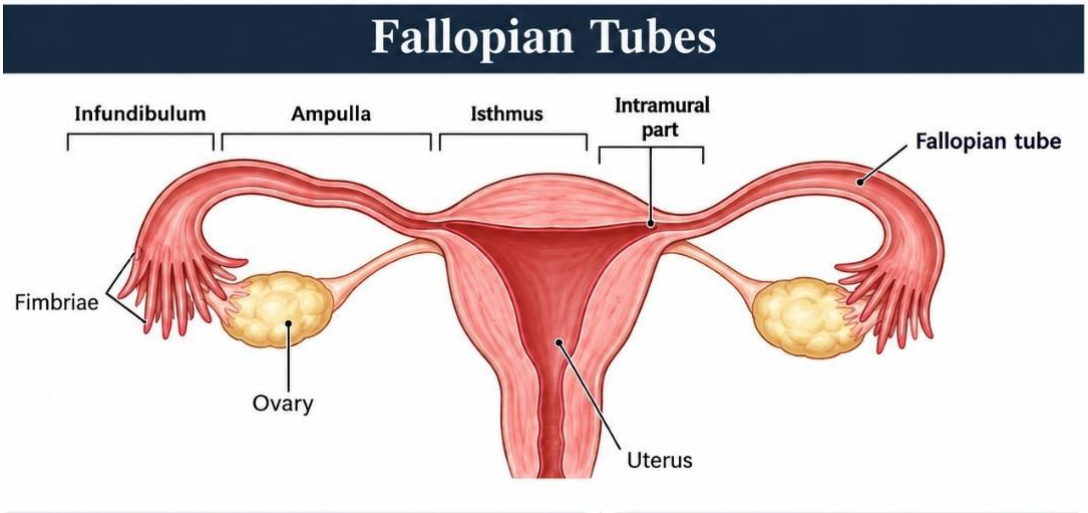
The fallopian tubes are a pair of narrow, muscular ducts about 10–12 cm long, extending from the uterus towards the ovaries.

1. Infundibulum: Funnel-shaped outer part that contains fimbriae.
2. Fimbriae: Finger-like projections that help capture the ovum during ovulation.
3. Ampulla: Widest part and the usual site of fertilization.
4. Isthmus: Narrow medial portion of the tube.
5. Intramural (Interstitial) part: Portion that passes through the uterine wall.

The wall consists of mucosa lined by ciliated epithelium, a muscular layer and an outer serosa.

- **Functions**

1. Ovum capture: Fimbriae help collect the ovum released from the ovary.
2. Fertilization: Provides the usual site for fertilization in the ampulla.
3. Transport: Cilia and muscular contractions move the ovum or fertilized egg towards the uterus.



20. Describe the major nutritional factors involved in the epigenetic regulation of foetal development. [U]/[CO2]

- Folic Acid and Vitamins: Support DNA methylation and gene regulation, essential for normal foetal growth and development.
- Proteins: Provide amino acids required for cell growth, tissue formation and epigenetic regulation.
- Iron and Iodine: Support brain development, nervous system function and normal cognitive development.
- Omega-3 Fatty Acids: Promote brain and nervous system development and influence gene expression.
- Antioxidants: Vitamins and minerals protect cells from oxidative damage and support normal foetal development. (2 marks in each point)